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COMMERCIAL ITEM DESCRIPTION

DETERGENT, LABORATORY GLASSWARE

The General Services Administration has authorized the use of this commercial item description in lieu of Federal Specification P-D-1526.

This commercial item description covers powdered or granular detergents for use in cleaning laboratory glassware and surgical instruments.

Salient characteristics:

The detergent shall be powdered or granular in form, and free flowing without caking. The detergent shall be formulated to remove biological and chemical soils from glassware and stainless steel. The detergent shall be completely soluble in water.

Cleaning efficiency. Under the following conditions, the detergent shall be able to clean a minimum of four out of five of the test beakers. Prepare 500 ml of a 1 percent solution of the test cleaning compound in hard water containing 0.264 mg/liter $CaCl_2 \cdot 2H_2O$ and 0.295 mg/liter $MgSO_4 \cdot 7H_2O$. Using a kitchen blender, emulsify 40 gm fresh beef liver, 20 gm fresh eggwhite, 10 gm of hydrogenated vegetable oil, and 80 ml of distilled water. Pour the emulsified soil into five clean 250 ml beakers and pour out the soil while coating the entire interior of each beaker as uniformly as possible. In order to soil the beakers efficiently, pour all the soil from one beaker to the next, until all five are soiled. The test beakers shall be sufficiently clean before soiling so that when rinsed with water, beads of water shall not form on the surface. Support each soiled beaker in an inverted position on two glass stirring rods and allow to dry overnight. Set each beaker upright and pour 100 ml of the test cleaner (at $20^{\circ} \pm 20^{\circ}$C) into each one without allowing the liquid to touch the sides above the 100 ml mark. Let stand 30 minutes at $20^{\circ} \pm 2^{\circ}$ C. Decant the solution and rinse twice with 100 ml portions of distilled water and allow to dry in an inverted position. Examine the bottom of each beaker for streaks, or soiled areas. A beaker showing any streaks or residues shall be considered not cleaned.

Rinsibility. Under the following conditions, the detergent shall not leave any detectable residue, or allow water droplet formation. Place 5 gms of the detergent on a clean glass plate about four (4) inches by six (6) inches by 0.08 inches. Add sufficient water to make a thin paste. Spread this paste evenly on one side of the piece of glass. Then rinse the glass plate by shaking it for three 10-second periods in successive 2-liter beakers of distilled water at 190° F (88° C). Inspect the plate visually for any adherent cleaner. Hold the plate vertically 45 seconds, then inspect for the presence of water droplets on the surface.

Solubility. The detergent shall demonstrate complete solubility in hard water under the following conditions. Place one gram of the detergent in 100 ml of synthetic hard water at 25° C. Stir with a magnetic stirrer for 10 miutes. After this time there shall be no evidence of cloudiness or a residue. The hard water shall consist of: 0.264 gm/liter $CaCl_2 \cdot 2H_2O$ and 0.295 gm/liter $MgSO_4 \cdot 7H_2O$.

Hard water stability. A 0.25 percent solution of the detergent shall be able to dissolve without precipitates in 30-grain hard water as evidenced by the following test. Dissolve 2.27 gm $CaCl_2 \cdot 2.52$ gm $MgSO_4 \cdot 7H_2O$ in separate 100 ml volumetric flasks and fill to volume with distilled water. Pipet a 10 ml aliquot of each of these solutions into one 500 ml volumetric flask and fill to volume with distilled water. Place 2.5 grams of the detergent sample into a 500 ml volumetric flask, and fill to volume with distilled water. Place 50 ml of each solution into a beaker. Transfer an additional 50 ml aliquot from the 500 ml flask containing the detergent sample solution to a 100 ml volumetric flask and fill to the mark with distilled water. This is the reference solution. Pipette 10 ml of the solution from the beaker into a 20 mm by 150 mm test tube. Pipette 10 ml of the reference solution into an identical test tube. Heat the test tubes in a water bath at 66° C for 5 minutes and allow to cool for 30 minutes. There shall be no flock or precipitate formed in the hard water sample.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Packaging, packing, and marking. The packaging and packing shall be in accordance with normal commercial practice and shall assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with the National Motor Freight Classification and Uniform Freight Classification. Marking shall be as specified in the contract or order.

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